

TE Curve Verification Pipeline Multi-Model Curve Comparison Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative TE Curve Verification Pipeline TE curves by overlaying multiple model predictions on the same original measured curve. The plots are intended to show whether each model tracks the local harmonic oscillations rather than only the broad mean trend.

Scope

- each comparison image contains four deterministic held-out test curves;
- forward comparisons are shown on forward curves only;
- backward comparisons are shown on backward curves only;
- Wave 1 screening keeps the three strongest family-best models by Curve MAE [deg] within each direction;
- Original Curve uses the same visual weight as predictions and a dark-gray color for balanced comparison.

Metrics Summary

Backward Reference Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860

Backward Wave 1 Family Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_bw	wave1_current_registry	Bw	0.008079	0.008672	17.892
harmonic_regression_bw	wave1_current_registry	Bw	0.003678	0.004012	8.058
periodic_mlp_bw	wave1_current_registry	Bw	0.010754	0.011174	23.160
residual_harmonic_mlp_bw	wave1_current_registry	Bw	0.008619	0.009001	18.775
tree_bw	wave1_current_registry	Bw	0.015215	0.015523	34.609
periodic_mlp_harmonic_bw	wave1_periodic_mlp_harmonic_campaign	Bw	0.003583	0.003925	7.875

Backward Wave 2.1 Temporal Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.008956	0.009381	19.524
gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.007468	0.007885	16.367
lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.006945	0.007367	15.245
periodic_temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.007596	0.008027	16.602
periodic_gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.006320	0.006795	14.078
periodic_lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.010769	0.011111	23.306

Backward Wave 2.3 Residual Harmonic Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
residual_harmonic_gru_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003502	0.003857	7.654
residual_harmonic_gru_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.008984	0.012987	20.358
residual_harmonic_gru_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.009370	0.013165	21.267
residual_harmonic_lstm_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003440	0.003793	7.510
residual_harmonic_lstm_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.007367	0.009945	16.660
residual_harmonic_lstm_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.010268	0.014769	23.355

Backward Wave 3.1 Offset-Aware Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_1_sequential_residual_offset_probe_bw	wave3_1_offset_aware_probe_registry	Bw	0.003636	0.004065	7.952

Backward Wave 3.2 Harmonic-Offset Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_2_clean_sequential_residual_offset_bw	wave3_2_harmonic_offset_probe_registry	Bw	0.003541	0.003971	7.732
wave3_2_harmonic_residual_offset_bw	wave3_2_harmonic_offset_probe_registry	Bw	0.003331	0.003671	7.261

Backward Wave 3.3 Curve-Aware Training Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_3_curve_aware_pointwise_control_bw	wave3_3_curve_aware_training_registry	Bw	0.003436	0.003761	7.538
wave3_3_raw_centered_shape_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.003465	0.003790	7.582
wave3_3_raw_offset_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.003469	0.003799	7.608
wave3_3_full_curve_composite_bw	wave3_3_curve_aware_training_registry	Bw	0.003510	0.003897	7.683

Backward Wave 4.1 Robust-Loss Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_1_mae_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003433	0.003750	7.506
wave4_1_smooth_l1_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003078	0.003403	6.676
wave4_1_log_cosh_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003486	0.003811	7.628

Backward Wave 4.2 Quantile Probabilistic Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_2_quantile_p10_p50_p90_bw	wave4_2_probabilistic_registry	Bw	0.002935	0.003250	6.307
wave4_2_gaussian_nll_bw	wave4_2_probabilistic_registry	Bw	0.003001	0.003303	6.488

Backward Wave 4.3 Mixture Density Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_3_mixture_density_k2_bw	wave4_3_mixture_density_registry	Bw	0.002668	0.002947	5.880
wave4_3_mixture_density_k3_bw	wave4_3_mixture_density_registry	Bw	0.002730	0.003009	6.049

Backward Wave 4.4 Latent State Hysteresis Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_4_gru_latent_offset_residual_bw	wave4_4_latent_state_hysteresis_registry	Bw	0.003542	0.003984	7.736
wave4_4_causal_tcn_latent_offset_residual_bw	wave4_4_latent_state_hysteresis_registry	Bw	0.003624	0.004098	7.903

Backward Wave 3 Harmonic Prior Residual Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave5_1_harmonic_prior_pointwise_control_bw	wave5_1_harmonic_prior_residual_registry	Bw	0.003360	0.003677	7.363
wave5_1_harmonic_prior_smooth_l1_structured_bw	wave5_1_harmonic_prior_residual_registry	Bw	0.003431	0.003739	7.523

Backward Wave52b Offset Harmonic Guided Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52b_offset_centered_shape_harmonic_Bw	wave52b_offset_harmonic_guided_registry	Bw	0.020690	0.021139	48.162

Backward Polished Model Development Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
polished_feedforward_Bw	polished_model_development_registry	Bw	0.008079	0.008672	17.892
polished_harmonic_regression_Bw	polished_model_development_registry	Bw	0.003678	0.004012	8.058
polished_periodic_mlp_Bw	polished_model_development_registry	Bw	0.010754	0.011174	23.160
polished_residual_harmonic_mlp_Bw	polished_model_development_registry	Bw	0.008619	0.009001	18.775
polished_tree_Bw	polished_model_development_registry	Bw	0.015215	0.015523	34.609
polished_periodic_mlp_harmonic_Bw	polished_model_development_registry	Bw	0.009393	0.009831	20.198
polished_temporal_convolution_Bw	polished_model_development_registry	Bw	0.008956	0.009381	19.524
polished_gru_sequence_Bw	polished_model_development_registry	Bw	0.007468	0.007885	16.367
polished_lstm_sequence_Bw	polished_model_development_registry	Bw	0.006945	0.007367	15.245
polished_periodic_temporal_convolution_Bw	polished_model_development_registry	Bw	0.007596	0.008027	16.602
polished_periodic_gru_sequence_Bw	polished_model_development_registry	Bw	0.006320	0.006795	14.078
polished_periodic_lstm_sequence_Bw	polished_model_development_registry	Bw	0.010769	0.011111	23.306
polished_residual_harmonic_gru_sequence_sparse_rcim_Bw	polished_model_development_registry	Bw	0.007012	0.007383	15.369
polished_residual_harmonic_gru_sequence_dense240_Bw	polished_model_development_registry	Bw	0.009235	0.009959	20.204
polished_residual_harmonic_gru_sequence_dense360_Bw	polished_model_development_registry	Bw	0.010578	0.012666	23.297
polished_residual_harmonic_lstm_sequence_sparse_rcim_Bw	polished_model_development_registry	Bw	0.007740	0.008099	16.813
polished_residual_harmonic_lstm_sequence_dense240_Bw	polished_model_development_registry	Bw	0.006772	0.007632	14.902
polished_residual_harmonic_lstm_sequence_dense360_Bw	polished_model_development_registry	Bw	0.009108	0.011272	20.181
polished_wave3_1_sequential_residual_offset_probe_Bw	polished_model_development_registry	Bw	0.009712	0.010105	21.306
polished_wave3_2_clean_sequential_residual_offset_Bw	polished_model_development_registry	Bw	0.008107	0.008493	17.879
polished_wave3_2_harmonic_residual_offset_Bw	polished_model_development_registry	Bw	0.004950	0.005267	11.001
polished_wave3_3_curve_aware_pointwise_control_Bw	polished_model_development_registry	Bw	0.007181	0.007525	15.776
polished_wave3_3_raw_centered_shape_curve_aware_Bw	polished_model_development_registry	Bw	0.005377	0.005696	11.898
polished_wave3_3_raw_offset_curve_aware_Bw	polished_model_development_registry	Bw	0.007815	0.008139	17.245
polished_wave3_3_full_curve_composite_Bw	polished_model_development_registry	Bw	0.005651	0.006040	12.391
polished_wave4_1_mae_robust_loss_Bw	polished_model_development_registry	Bw	0.008153	0.008484	17.810
polished_wave4_1_smooth_l1_robust_loss_Bw	polished_model_development_registry	Bw	0.007174	0.007502	15.817
polished_wave4_1_log_cosh_robust_loss_Bw	polished_model_development_registry	Bw	0.004849	0.005168	10.664
polished_wave4_2_quantile_p10_p50_p90_Bw	polished_model_development_registry	Bw	0.003809	0.004175	8.347
polished_wave4_2_gaussian_nll_Bw	polished_model_development_registry	Bw	0.005195	0.005557	11.506
polished_wave4_3_mixture_density_k2_Bw	polished_model_development_registry	Bw	0.004718	0.005044	10.385
polished_wave4_3_mixture_density_k3_Bw	polished_model_development_registry	Bw	0.005049	0.005359	11.212

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
polished_wave4_4_gru_latent_offset_residual_Bw	polished_model_development_registry	Bw	0.008062	0.008539	17.742
polished_wave4_4_causal_tcn_latent_offset_residual_Bw	polished_model_development_registry	Bw	0.007758	0.008220	17.080
polished_wave5_1_harmonic_prior_pointwise_control_Bw	polished_model_development_registry	Bw	0.007570	0.007930	16.593
polished_wave5_1_harmonic_prior_smooth_l1_structured_Bw	polished_model_development_registry	Bw	0.006945	0.007295	15.256

Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
periodic_mlp_harmonic_bw	wave1_periodic_mlp_harmonic_campaign	Bw	0.003583	0.003925	7.875
harmonic_regression_bw	wave1_current_registry	Bw	0.003678	0.004012	8.058
feedforward_bw	wave1_current_registry	Bw	0.008079	0.008672	17.892

Backward Reference Tree And Wave 2.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.015215	0.015523	34.609
temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.008956	0.009381	19.524
gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.007468	0.007885	16.367
lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.006945	0.007367	15.245
periodic_temporal_convolution_bw	wave2_temporal_entry_registry	Bw	0.007596	0.008027	16.602
periodic_gru_sequence_bw	wave2_temporal_entry_registry	Bw	0.006320	0.006795	14.078
periodic_lstm_sequence_bw	wave2_temporal_entry_registry	Bw	0.010769	0.011111	23.306

Backward Reference Tree And Wave 2.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.015215	0.015523	34.609
residual_harmonic_gru_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003502	0.003857	7.654
residual_harmonic_gru_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.008984	0.012987	20.358
residual_harmonic_gru_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.009370	0.013165	21.267
residual_harmonic_lstm_sequence_sparse_rcim_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.003440	0.003793	7.510
residual_harmonic_lstm_sequence_dense240_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.007367	0.009945	16.660
residual_harmonic_lstm_sequence_dense360_Bw	wave2c_residual_harmonic_temporal_registry	Bw	0.010268	0.014769	23.355

Backward Reference Tree And Wave 3.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.015215	0.015523	34.609
wave3_1_sequential_residual_offset_probe_bw	wave3_1_offset_aware_probe_registry	Bw	0.003636	0.004065	7.952

Backward Reference Tree And Wave 3.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.015215	0.015523	34.609
wave3_3_curve_aware_pointwise_control_bw	wave3_3_curve_aware_training_registry	Bw	0.003436	0.003761	7.538
wave3_3_raw_centered_shape_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.003465	0.003790	7.582
wave3_3_raw_offset_curve_aware_bw	wave3_3_curve_aware_training_registry	Bw	0.003469	0.003799	7.608
wave3_3_full_curve_composite_bw	wave3_3_curve_aware_training_registry	Bw	0.003510	0.003897	7.683

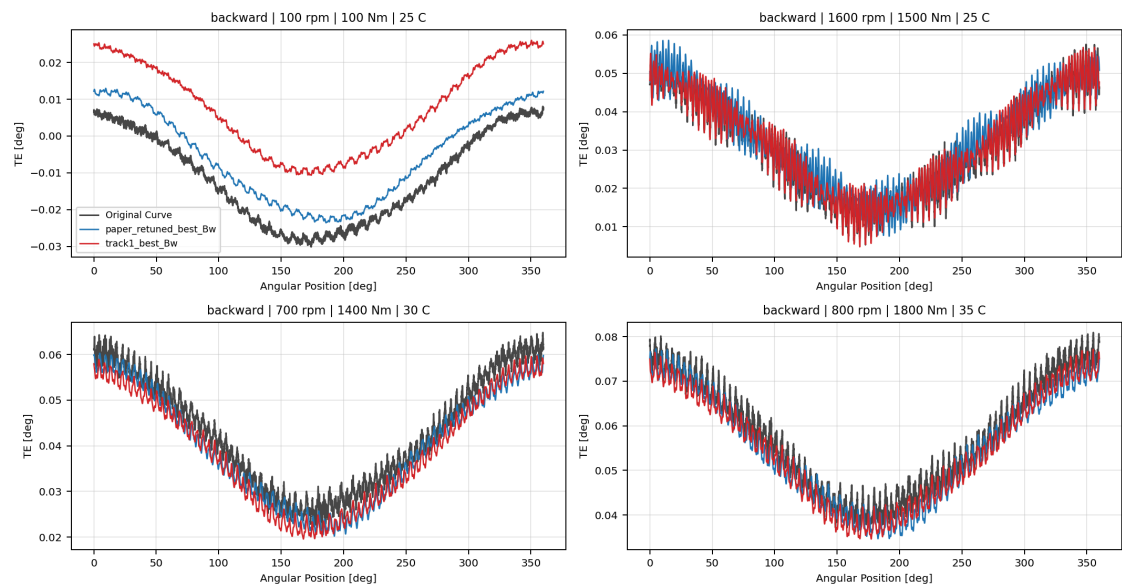
Backward Reference Tree And Wave 4 series Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Bw	rcim_retuned	Bw	0.003675	0.004284	7.572
rcim_model_bank_reproduction_best_bw	rcim_model_bank_reproduction	Bw	0.005027	0.005212	11.860
tree_bw	wave1_current_registry	Bw	0.015215	0.015523	34.609
wave4_1_mae_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003433	0.003750	7.506
wave4_1_smooth_l1_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003078	0.003403	6.676
wave4_1_log_cosh_robust_loss_bw	wave4_1_robust_loss_registry	Bw	0.003486	0.003811	7.628

Comparison Gallery - Backward Reference Model Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` .

Backward Reference Model Overlay

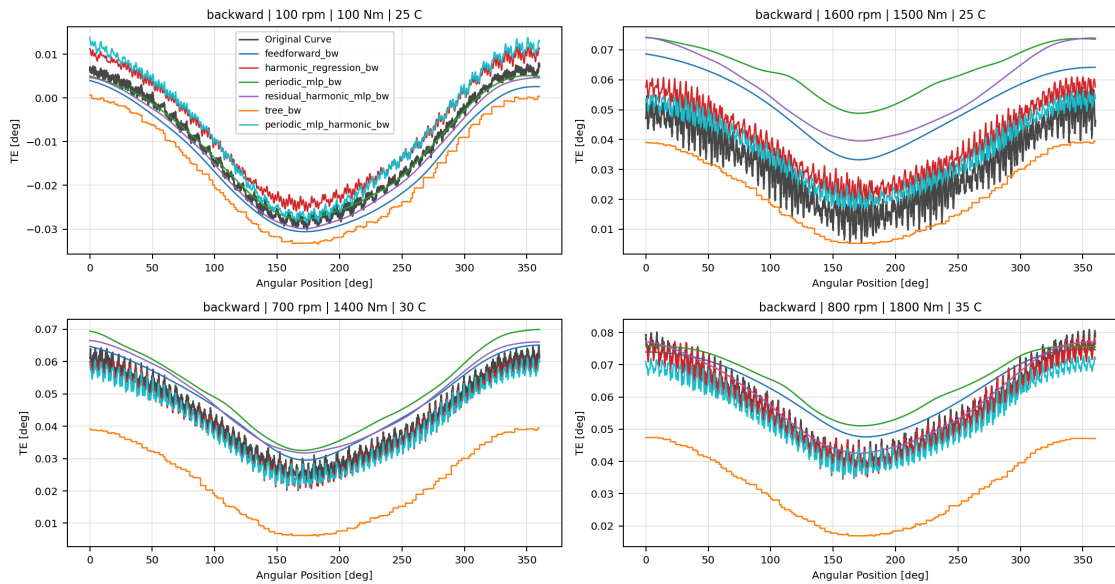


Backward Reference Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 1 Family Model Overlay

Included models: `feedforward_bw` , `harmonic_regression_bw` , `periodic_mlp_bw` , `residual_harmonic_mlp_bw` , `tree_bw` , `periodic_mlp_harmonic_bw` .

Backward Wave 1 Family Model Overlay

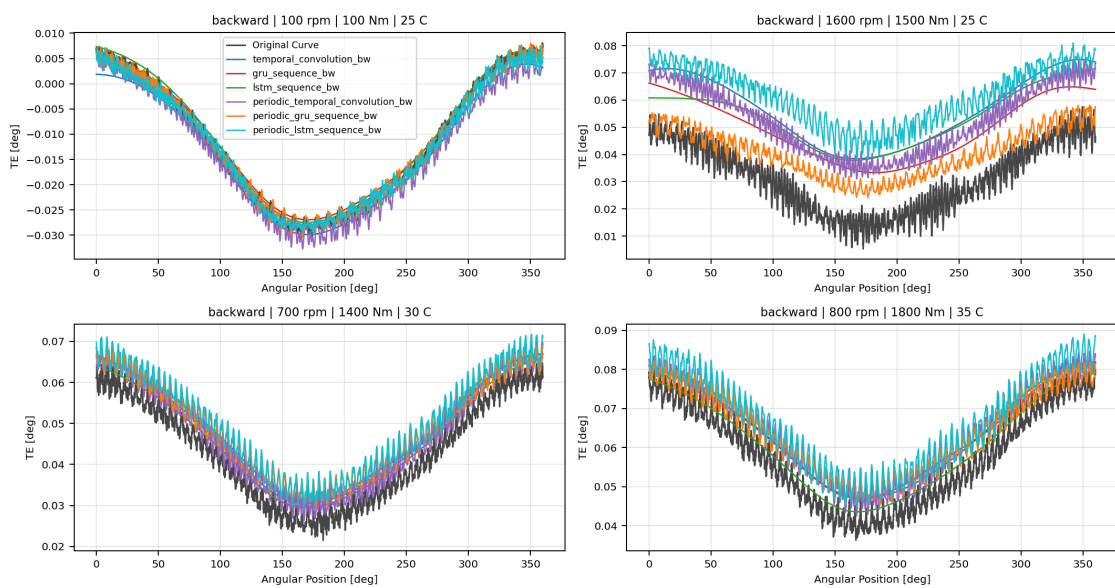


Backward Wave 1 Family Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 2.1 Temporal Model Overlay

Included models: `temporal_convolution_bw` , `gru_sequence_bw` , `lstm_sequence_bw` , `periodic_temporal_convolution_bw` , `periodic_gru_sequence_bw` , `periodic_lstm_sequence_bw` .

Backward Wave 2.1 Temporal Model Overlay

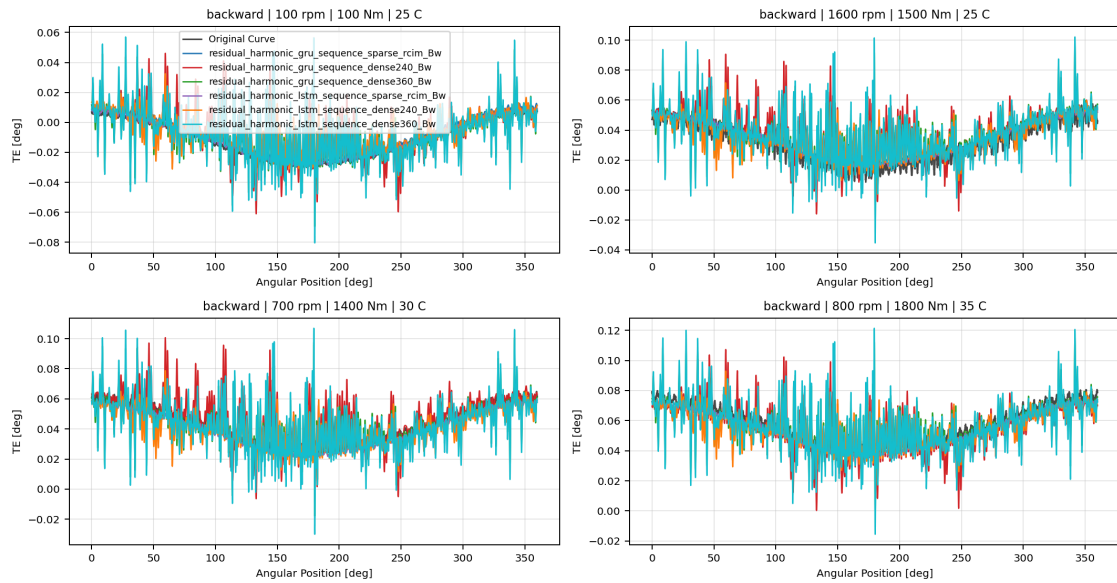


Backward Wave 2.1 Temporal Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 2.3 Residual Harmonic Temporal Overlay

Included models: `residual_harmonic_gru_sequence_sparse_rcim_Bw` ,
`residual_harmonic_gru_sequence_dense240_Bw` , `residual_harmonic_gru_sequence_dense360_Bw` ,
`residual_harmonic_lstm_sequence_sparse_rcim_Bw` , `residual_harmonic_lstm_sequence_dense240_Bw` ,
`residual_harmonic_lstm_sequence_dense360_Bw` .

Backward Wave 2.3 Residual Harmonic Temporal Overlay

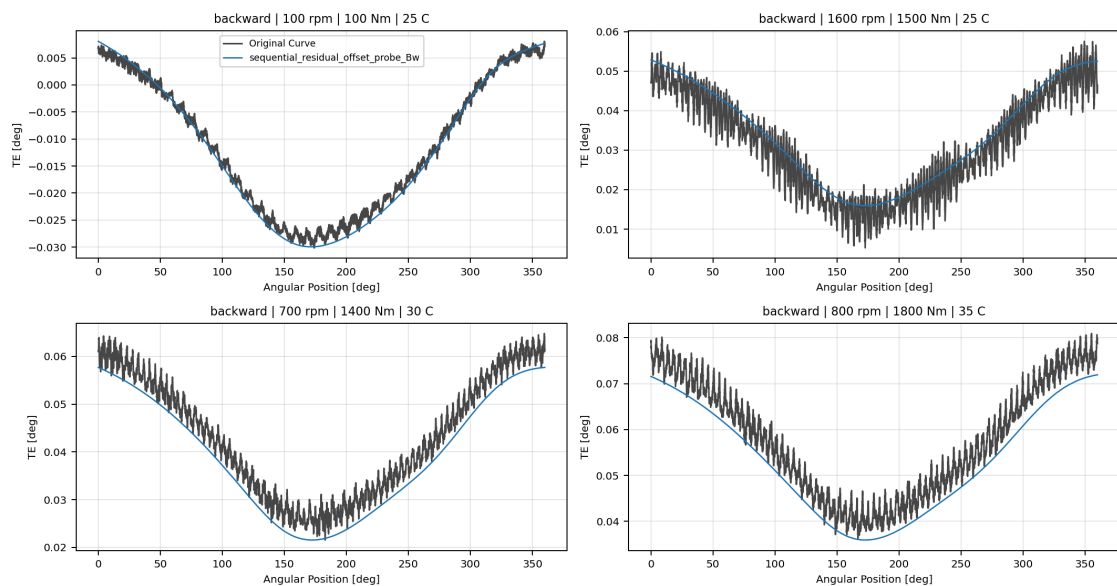


Backward Wave 2.3 Residual Harmonic Temporal Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 3.1 Offset-Aware Probe Overlay

Included models: `wave3_1_sequential_residual_offset_probe_bw` .

Backward Wave 3.1 Offset-Aware Probe Overlay

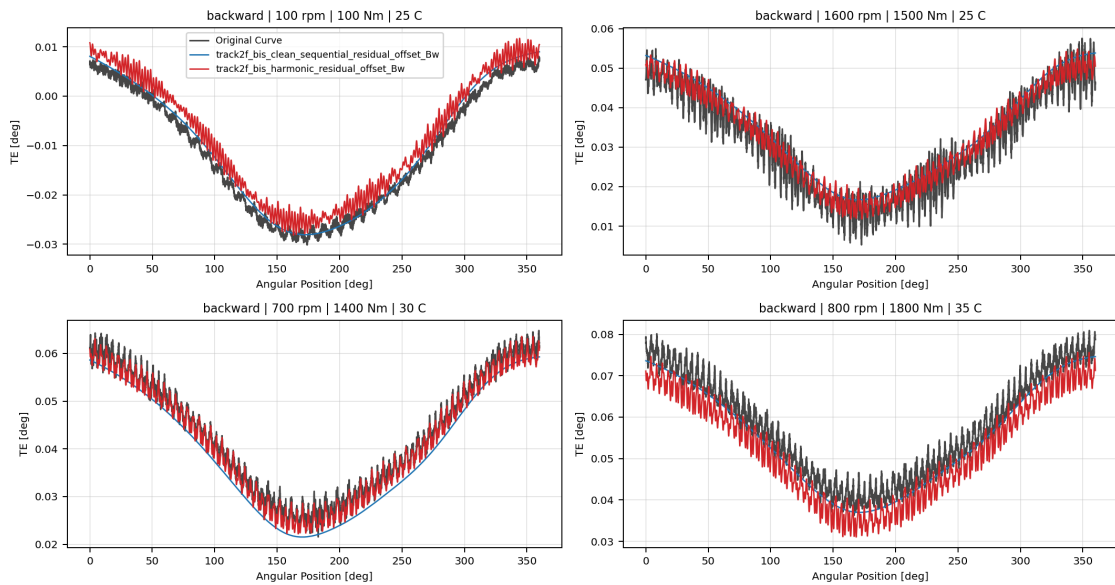


Backward Wave 3.1 Offset-Aware Probe Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 3.2 Harmonic-Offset Probe Overlay

Included models: `wave3_2_clean_sequential_residual_offset_bw` , `wave3_2_harmonic_residual_offset_bw` .

Backward Wave 3.2 Harmonic-Offset Probe Overlay

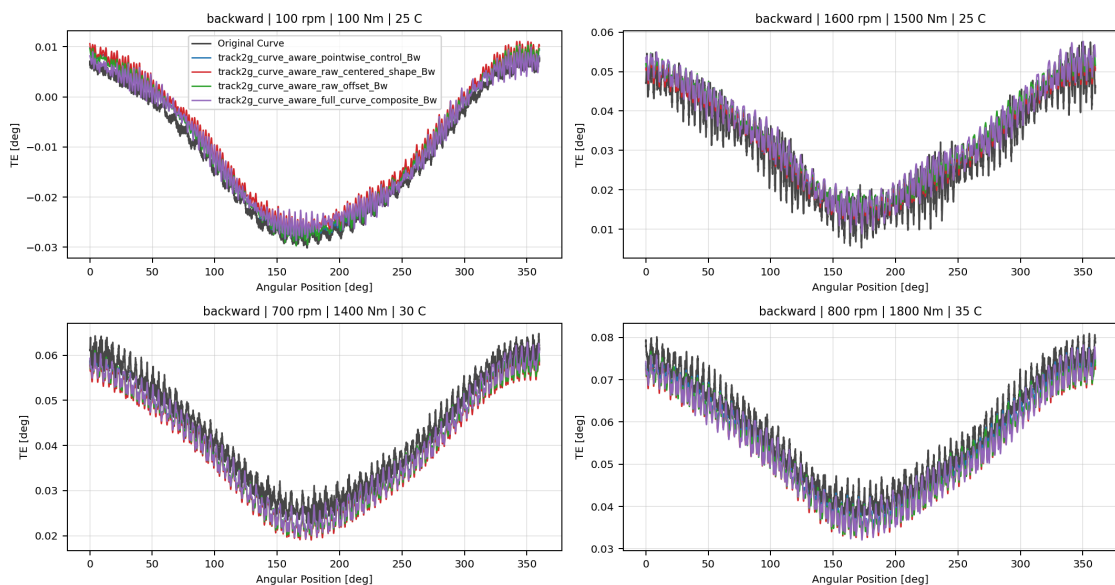


Backward Wave 3.2 Harmonic-Offset Probe Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 3.3 Curve-Aware Training Overlay

Included models: `wave3_3_curve_aware_pointwise_control_bw` , `wave3_3_raw_centered_shape_curve_aware_bw` , `wave3_3_raw_offset_curve_aware_bw` , `wave3_3_full_curve_composite_bw` .

Backward Wave 3.3 Curve-Aware Training Overlay

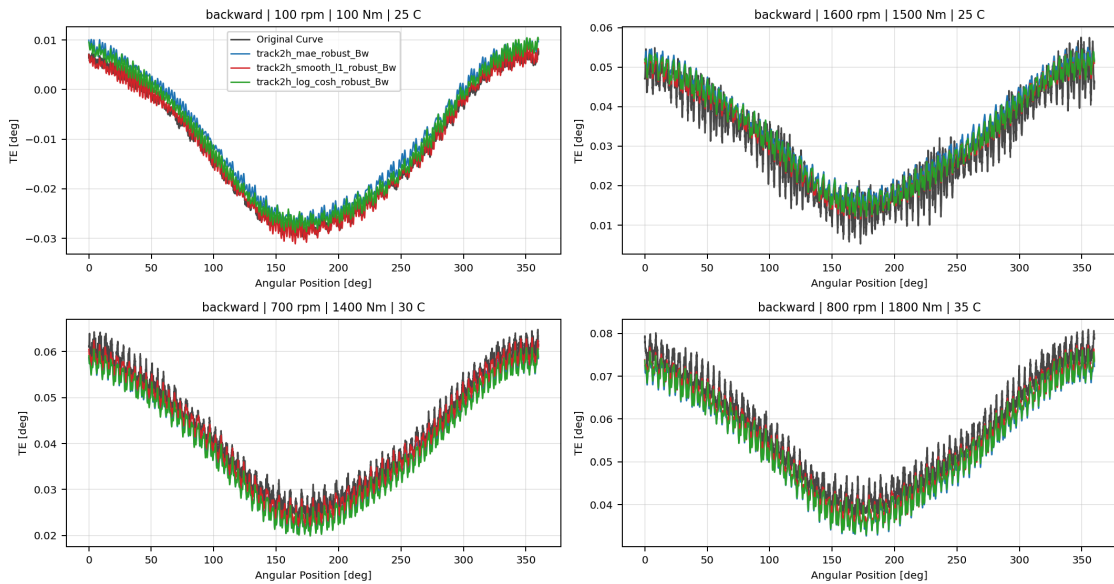


Backward Wave 3.3 Curve-Aware Training Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 4.1 Robust-Loss Overlay

Included models: `wave4_1_mae_robust_loss_bw` , `wave4_1_smooth_l1_robust_loss_bw` ,
`wave4_1_log_cosh_robust_loss_bw` .

Backward Wave 4.1 Robust-Loss Overlay

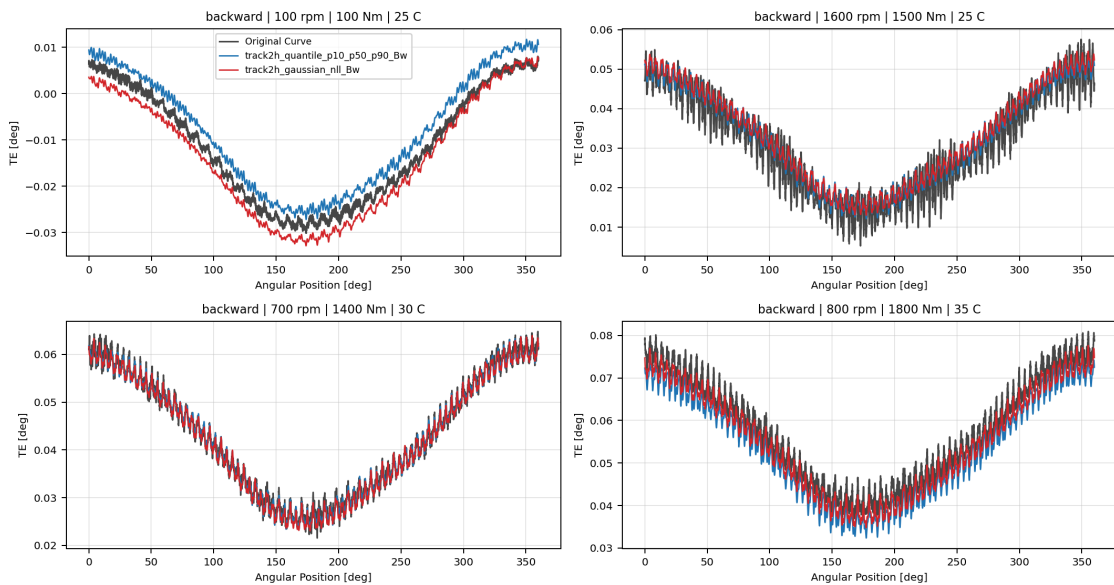


Backward Wave 4.1 Robust-Loss Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 4.2 Quantile Probabilistic Overlay

Included models: `wave4_2_quantile_p10_p50_p90_bw` , `wave4_2_gaussian_nll_bw` .

Backward Wave 4.2 Quantile Probabilistic Overlay

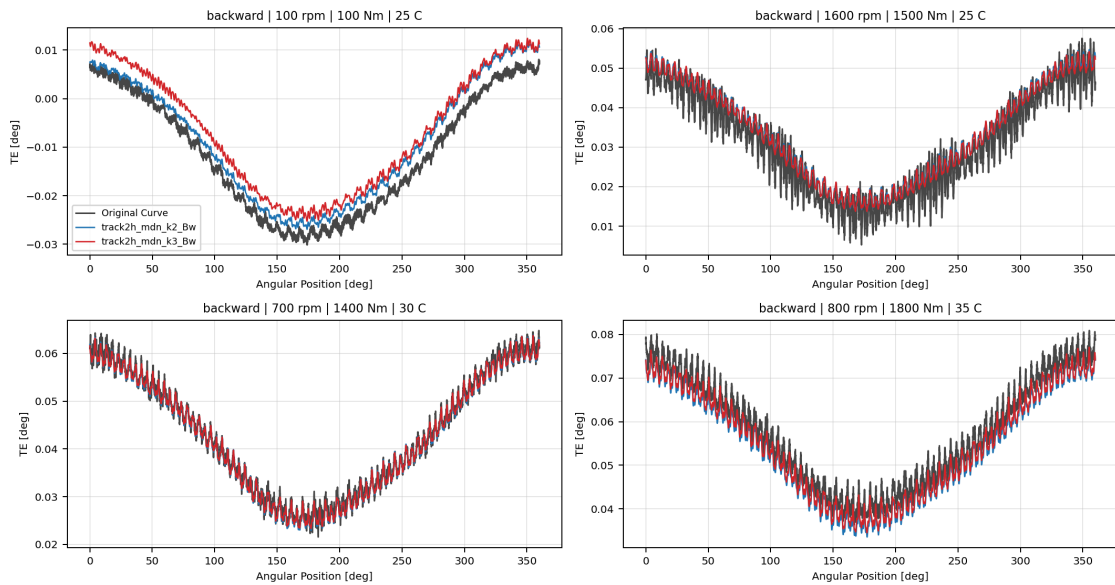


Backward Wave 4.2 Quantile Probabilistic Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 4.3 Mixture Density Overlay

Included models: `wave4_3_mixture_density_k2_bw` , `wave4_3_mixture_density_k3_bw` .

Backward Track2h Mixture Density Heads Registry Overlay

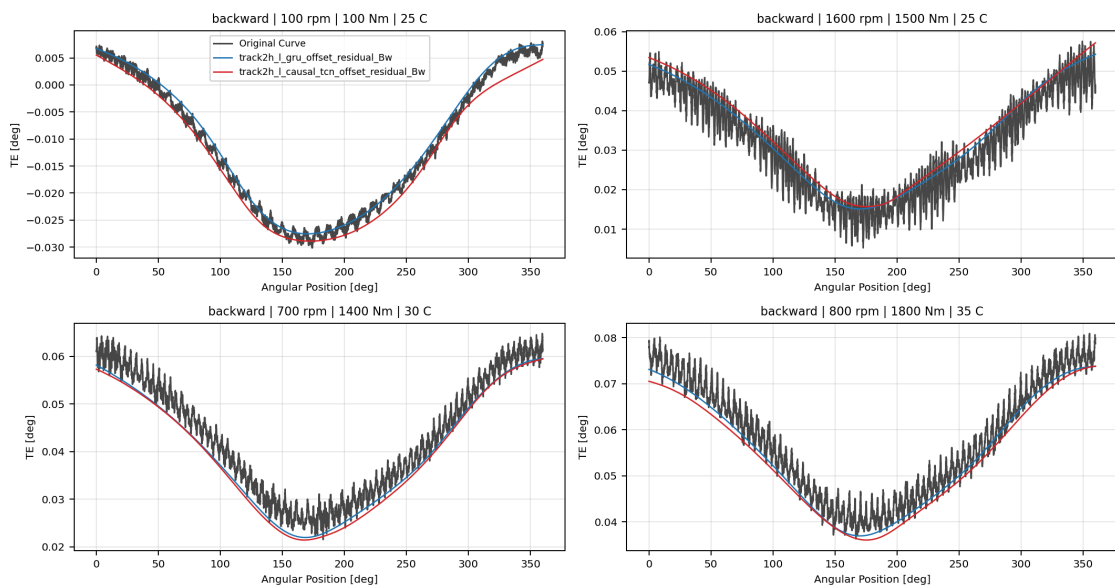


Backward Wave 4.3 Mixture Density Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 4.4 Latent State Hysteresis Overlay

Included models: `wave4_4_gru_latent_offset_residual_bw` , `wave4_4_causal_tcn_latent_offset_residual_bw` .

Backward Track2h Latent State Hysteresis Registry Overlay

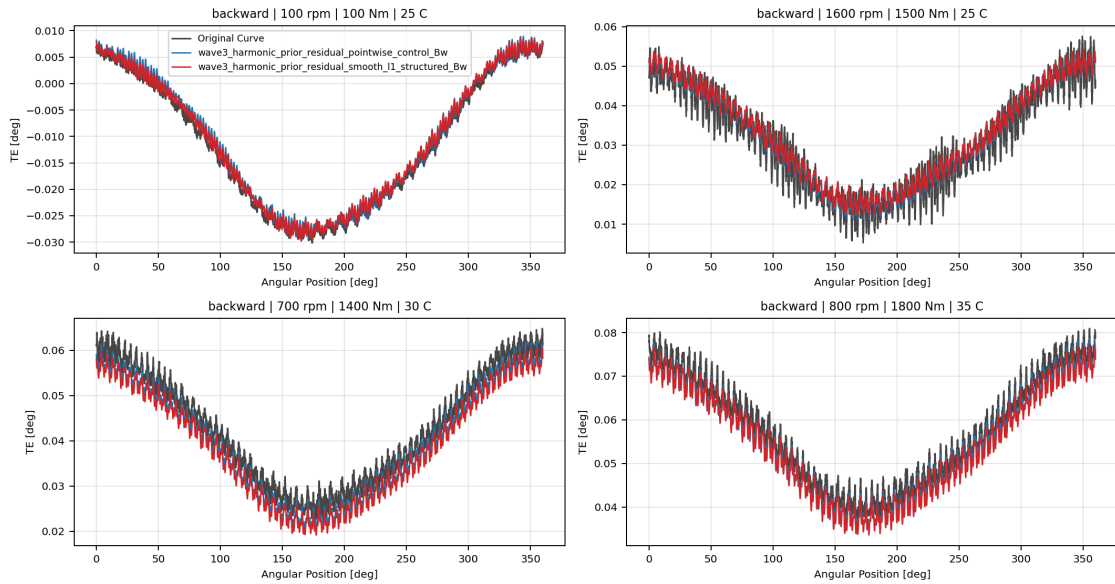


Backward Wave 4.4 Latent State Hysteresis Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave 3 Harmonic Prior Residual Registry Overlay

Included models: `wave5_1_harmonic_prior_pointwise_control_bw` ,
`wave5_1_harmonic_prior_smooth_l1_structured_bw` .

Backward Wave 3 Harmonic Prior Residual Registry Overlay

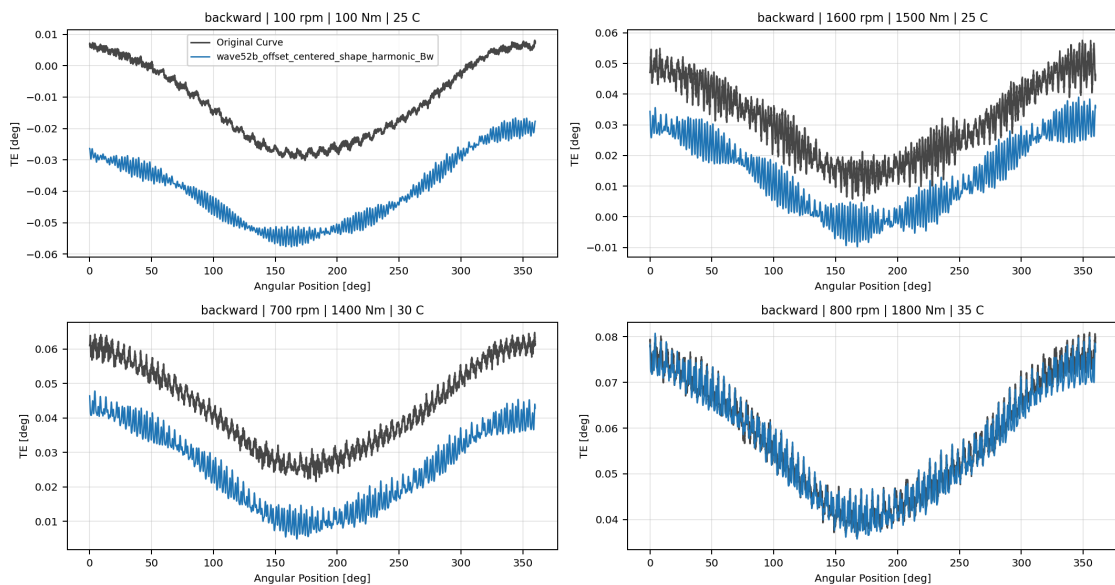


Backward Wave 3 Harmonic Prior Residual Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Wave52b Offset Harmonic Guided Registry Overlay

Included models: `wave52b_offset_centered_shape_harmonic_Bw` .

Backward Wave52b Offset Harmonic Guided Registry Overlay

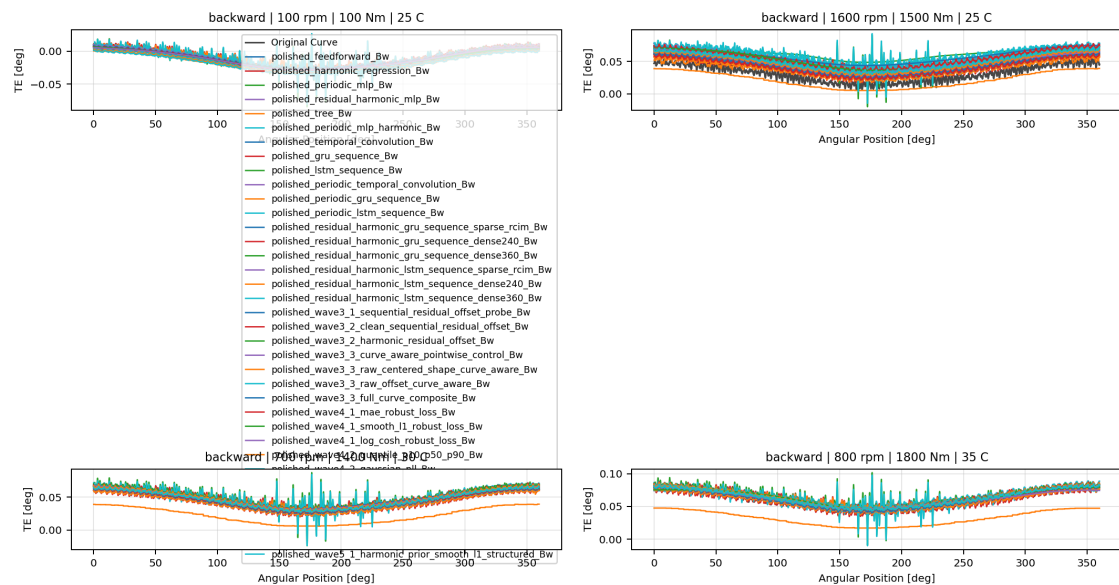


Backward Wave52b Offset Harmonic Guided Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Polished Model Development Registry Overlay

Included models: polished_feedforward_Bw , polished_harmonic_regression_Bw , polished_periodic_mlp_Bw , polished_residual_harmonic_mlp_Bw , polished_tree_Bw , polished_periodic_mlp_harmonic_Bw , polished_temporal_convolution_Bw , polished_gru_sequence_Bw , polished_lstm_sequence_Bw , polished_periodic_temporal_convolution_Bw , polished_periodic_gru_sequence_Bw , polished_periodic_lstm_sequence_Bw , polished_residual_harmonic_gru_sequence_sparse_rcim_Bw , polished_residual_harmonic_gru_sequence_dense240_Bw , polished_residual_harmonic_gru_sequence_dense360_Bw , polished_residual_harmonic_lstm_sequence_sparse_rcim_Bw , polished_residual_harmonic_lstm_sequence_dense240_Bw , polished_residual_harmonic_lstm_sequence_dense360_Bw , polished_wave3_1_sequential_residual_offset_probe_Bw , polished_wave3_2_clean_sequential_residual_offset_Bw , polished_wave3_2_harmonic_residual_offset_Bw , polished_wave3_3_curve_aware_pointwise_control_Bw , polished_wave3_3_raw_centered_shape_curve_aware_Bw , polished_wave3_3_raw_offset_curve_aware_Bw , polished_wave3_3_full_curve_composite_Bw , polished_wave4_1_mae_robust_loss_Bw , polished_wave4_1_smooth_l1_robust_loss_Bw , polished_wave4_1_log_cosh_robust_loss_Bw , polished_wave4_2_quantile_p10_p50_p90_Bw , polished_wave4_2_gaussian_nll_Bw , polished_wave4_3_mixture_density_k2_Bw , polished_wave4_3_mixture_density_k3_Bw , polished_wave4_4_gru_latent_offset_residual_Bw , polished_wave4_4_causal_tcn_latent_offset_residual_Bw , polished_wave5_1_harmonic_prior_pointwise_control_Bw , polished_wave5_1_harmonic_prior_smooth_l1_structured_Bw .

Backward Polished Model Development Registry Overlay

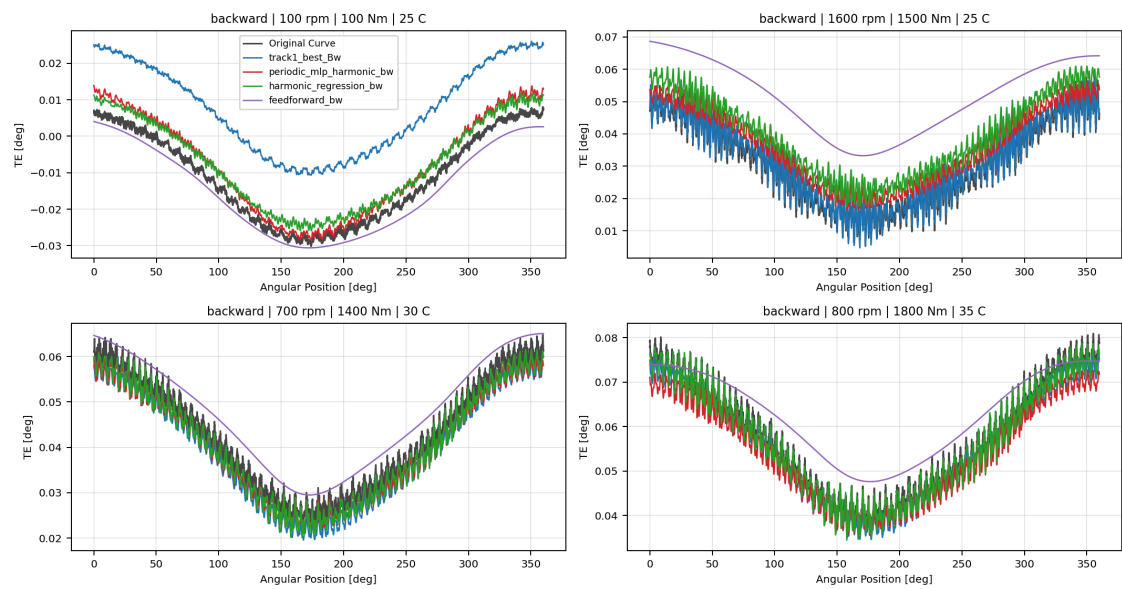


Backward Polished Model Development Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Included models: rcim_model_bank_reproduction_best_bw , periodic_mlp_harmonic_bw , harmonic_regression_bw , feedforward_bw .

Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

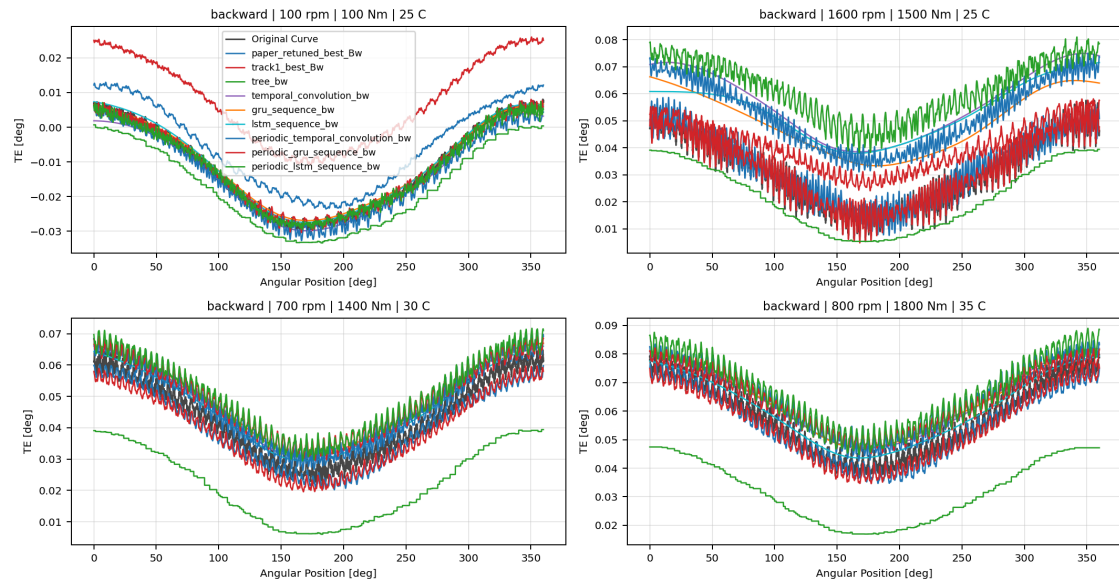


Backward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 2.1 Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` , `temporal_convolution_bw` , `gru_sequence_bw` , `lstm_sequence_bw` , `periodic_temporal_convolution_bw` , `periodic_gru_sequence_bw` , `periodic_lstm_sequence_bw` .

Backward Reference Tree And Wave 2.1 Overlay

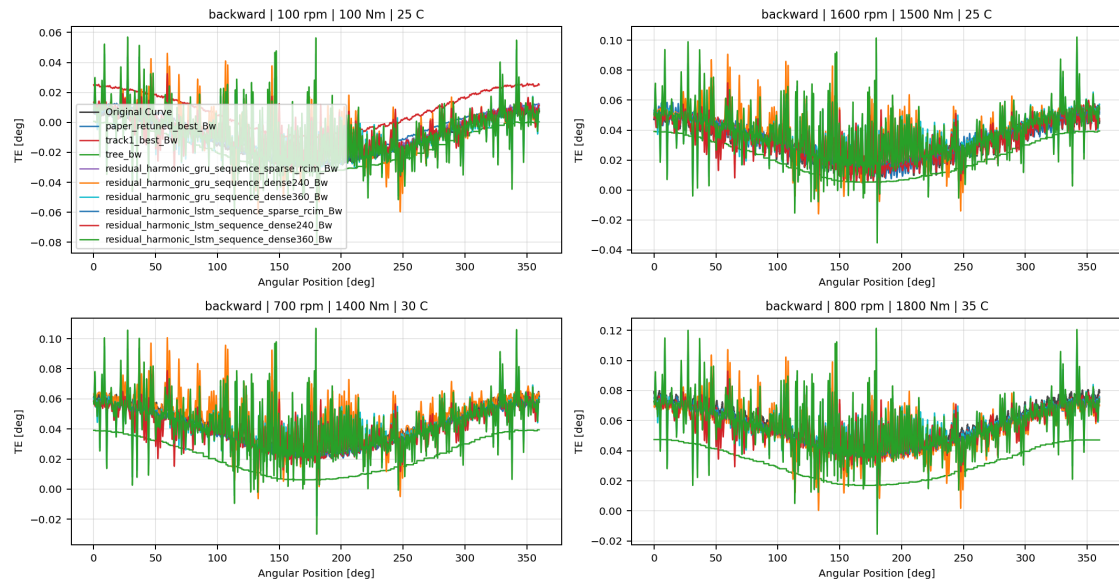


Backward Reference Tree And Wave 2.1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 2.3 Overlay

Included models: `paper_retuned_best_Bw`, `rcim_model_bank_reproduction_best_bw`, `tree_bw`, `residual_harmonic_gru_sequence_sparse_rcim_Bw`, `residual_harmonic_gru_sequence_dense240_Bw`, `residual_harmonic_gru_sequence_dense360_Bw`, `residual_harmonic_lstm_sequence_sparse_rcim_Bw`, `residual_harmonic_lstm_sequence_dense240_Bw`, `residual_harmonic_lstm_sequence_dense360_Bw`.

Backward Reference Tree And Wave 2.3 Overlay

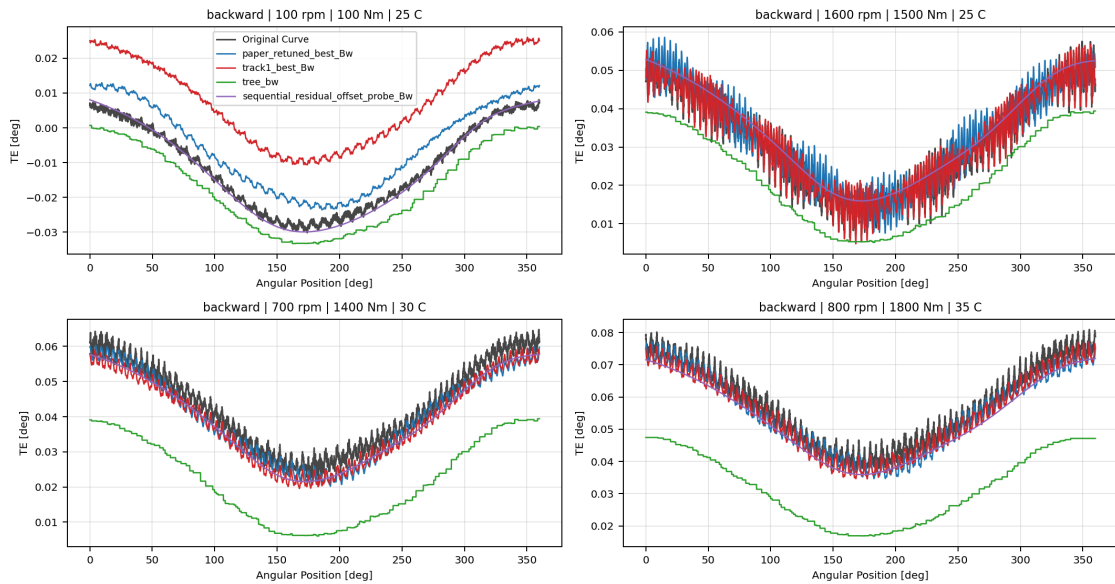


Backward Reference Tree And Wave 2.3 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 3.1 Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` , `wave3_1_sequential_residual_offset_probe_bw` .

Backward Reference Tree And Wave 3.1 Overlay

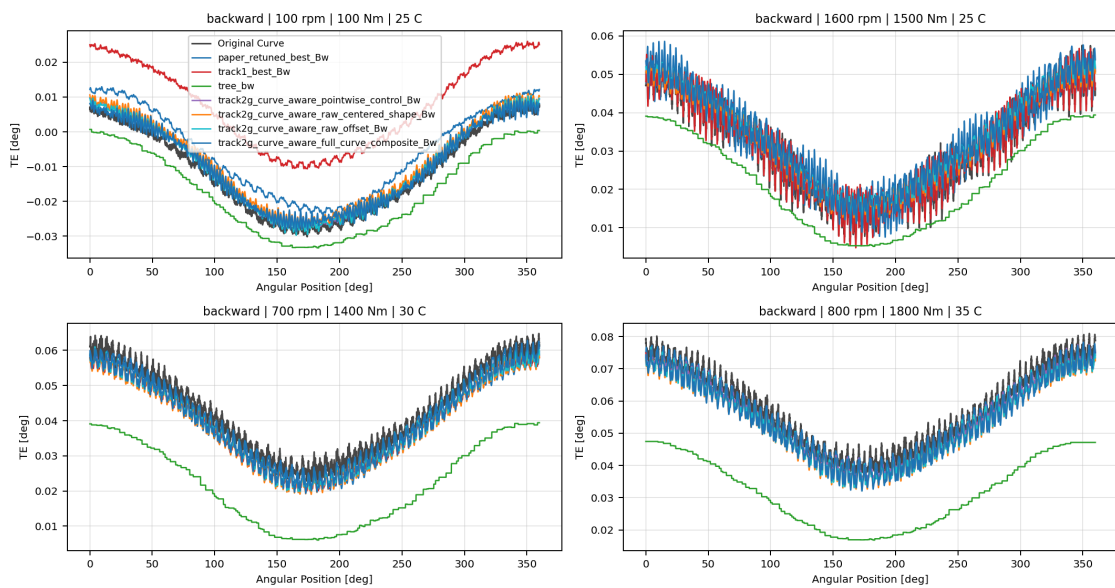


Backward Reference Tree And Wave 3.1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 3.3 Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` , `wave3_3_curve_aware_pointwise_control_bw` , `wave3_3_raw_centered_shape_curve_aware_bw` , `wave3_3_raw_offset_curve_aware_bw` , `wave3_3_full_curve_composite_bw` .

Backward Reference Tree And Wave 3.3 Overlay

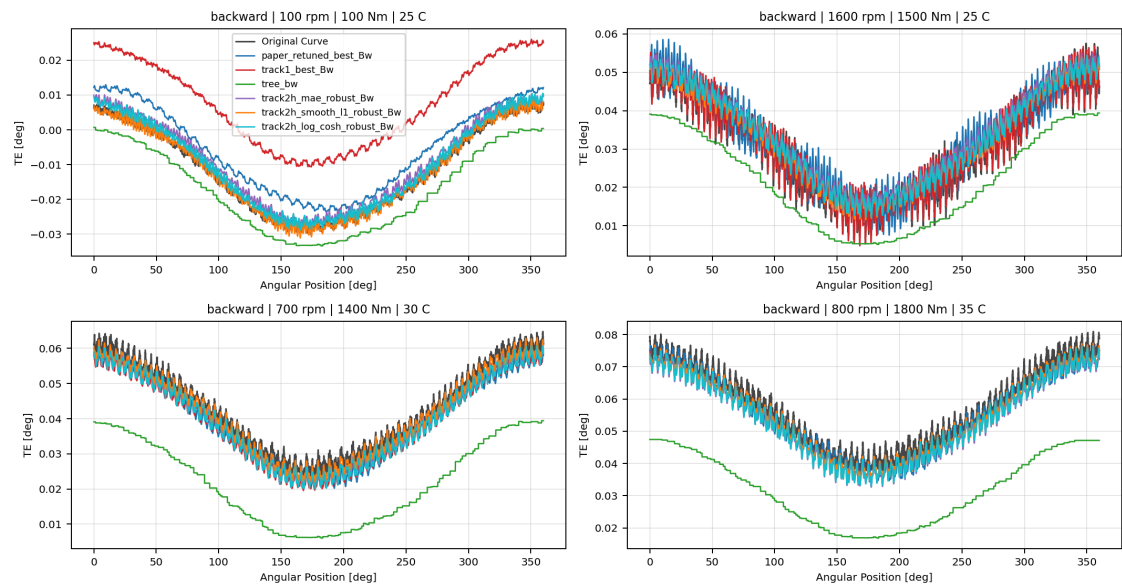


Backward Reference Tree And Wave 3.3 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Backward Reference Tree And Wave 4 series Overlay

Included models: `paper_retuned_best_Bw` , `rcim_model_bank_reproduction_best_bw` , `tree_bw` , `wave4_1_mae_robust_loss_bw` , `wave4_1_smooth_l1_robust_loss_bw` , `wave4_1_log_cosh_robust_loss_bw` .

Backward Reference Tree And Wave 4 series Overlay



Backward Reference Tree And Wave 4 series Overlay TE Curve Verification Pipeline comparison

Output Artifacts

- output directory: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-07-05-16-57-09__track2_multi_model_curve_comparison_report` ;
- summary YAML: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-07-05-16-57-09__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-07-05-16-57-09__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_metrics.csv` ;
- report Markdown:
`doc\reports\analysis\te_curve_verification_pipeline\02_visual_reports\dataset_surface_report\simplified_dataset\backward\overlay\[2026-07-04]\track2_multi_model_curve_comparison_report.md` .